Problem 2: You are given the value of n. Calculate the result of $1 + (2+3) + (4+5+6) + \cdots$ nth term. (Mahbubul. 32)

Solution:

$1 + (2+3) + (4+5+6) + \cdots$ nth term

From the series, we can see that the i-th term has i integers within it. Also, integers within the terms are unique and not duplicate of others. Furthermore, the integers are growing linearly, with difference of 1 from both the previous and next integer. Meaning it is a linear sequence of difference of 1.

Another observation, the last integer within the i-th term is actually the value of $\sum_{j=1}^{i} (j)$

$\therefore$ The last term n's last integer is, $\sum_{j=1}^{n} (j) = \frac{n(n+1)}{2}$

Applying the summation formula from 1 to the last integer, $\frac{n(n+1)}{2}$ we get,

$$\sum_{i=1}^{\frac{n(n+1)}{2}} (i) = \frac{\frac{n(n+1)}{2}\left(\frac{n(n+1)}{2}+1\right)}{2}$$

$$= \frac{\frac{n^2+n}{2} \quad \frac{n^2+n+2}{2}}{2}$$

$$= \frac{\frac{n^4+n^3+2n^2+n^3+n^2+2n}{4}}{2}$$

$$= \frac{n^4+2n^3+3n^2+2n}{8}$$

$$= \frac{n(n^3+2n^2+3n+2)}{8}$$